

Day 5 - Interview Questions

1. Explain the difference between positional arguments and keyword arguments in function calls.

Positional arguments are passed to a function based on their position in the function call. The order matters, and they are assigned to corresponding parameters in the function definition.

Keyword arguments are passed with parameter names in the function call. They allow you to specify which argument corresponds to which parameter, regardless of order.

2. What is a default argument in a Python function?

A default argument is a parameter in a function that has a default value assigned. If the argument is not provided when calling the function, it uses the default value. Default arguments are specified in the function definition.

3. What is a nested function in Python, and why might you use one?

A nested function is a function defined inside another function. It is useful for encapsulating functionality that is only relevant within the outer function, providing data hiding, and avoiding namespace pollution.

4. Explain the concept of "nonlocal" variables in nested functions.

"Nonlocal" is a keyword in Python that is used in a nested function to indicate that a variable should be treated as a variable in the nearest enclosing (non-local) scope, rather than creating a new local variable with the same name. This is useful when you want to modify a variable from an outer scope in a nested function.

5. What is the difference between global and local variables in Python, and how are they accessed within functions?

Global variables are defined outside of any function and can be accessed from any part of the code.

Local variables are defined inside a function and are only accessible within that function's scope.

6. What is the purpose of the str() function in Python, and when might you use it?

The str() function is used to convert a value to a string data type. You might use it when you want to represent non-string data as strings for display or when working with string operations.

7. What is type conversion in Python, and why is it important?

Type conversion, also known as type casting, is the process of converting one data type into another. It is important because it allows you to work with different data types, perform operations, and ensure data consistency in your programs.

8. Explain the purpose of the int(), float(), and str() functions in Python.

int(): It is used to convert a value to an integer data type.

float(): It is used to convert a value to a floating-point (decimal) data type.

str(): It is used to convert a value to a string data type.

9. How do you define a function in Python?

A function in Python is defined using the def keyword followed by the function name and parameters.

10. What is the purpose of the return statement in a Python function?

The return statement is used to specify the value that a function should return when it is called. It also terminates the function's execution. If no return statement is used, the function returns None by default.